Exp: 30 RTOS Task Scheduling

Program:

```
// Code your testbench here.
// Uncomment the next line for SystemC modules.
// #include <systemc>
#include <systemc.h>
SC_MODULE(rtos_scheduler) {
void task1() {
while(true) {
cout << "Task1 Running at "
<< sc_time_stamp() << endl;
wait(1, SC_SEC);
}
}
void task2() {
while(true) {
cout << "Task2 Running at "
<< sc_time_stamp() << endl;
wait(1, SC_SEC);
}
}
SC_CTOR(rtos_scheduler) {
SC_THREAD(task1);
SC_THREAD(task2);
}
};
int sc_main(int argc, char* argv[]) {
rtos_scheduler obj("Scheduler");
```

sc_start(6, SC_SEC);

return 0;

}

New Run Save Copy

testbench.cpp

```
// Code your testbench here.
// Uncomment the next line for SystemC modules.
// #include <systemc>
#include <systemc.h>
SC_MODULE(rtos_scheduler) {
void task1() {
while(true) {
cout << "Task1 Running at "
<< sc_time_stamp() << endl;
wait(1, SC_SEC);
}
}
void task2() {
while(true) {
cout << "Task2 Running at "
<< sc_time_stamp() << endl;
wait(1, SC_SEC);
}
}
SC_CTOR(rtos_scheduler) {
SC_THREAD(task1);
SC_THREAD(task2);
}
};
```

Output:

Log Share

```
Task2 Running at 0 s
Task1 Running at 1 s
Task2 Running at 1 s
Task1 Running at 2 s
Task2 Running at 2 s
Task1 Running at 3 s
Task2 Running at 3 s
Task1 Running at 4 s
Task2 Running at 4 s
Task1 Running at 5 s
Task2 Running at 5 s
Done
```